

```

In [20]: python_class={'Rahul','Priya','Arjun','Sneha','Karan'}
         java_class={'Priya','Arjun','Vikram','Neha'}
         ml_class={'Sneha','Vikram','Rahul','Divya'}
         b=[]
         p=[]
         u=set()
         o=0
         for i in python_class:
             if i in java_class:
                 b.append(i)
             if i not in java_class and i not in ml_class:
                 p.append(i)
         for i in python_class:
             for j in java_class:
                 for k in ml_class:
                     u.add(i)
                     u.add(j)
                     u.add(k)
         for i in u:
             if i in python_class and i not in java_class and i not in ml_class:
                 o+=1
             if i in java_class and i not in ml_class and i not in python_class:
                 o+=1
             if i in ml_class and i not in java_class and i not in python_class:
                 o+=1
         print('Python & Java :',b)
         print('Python only :',p)
         print('All unique :',u)
         print('Only one course:',o)

```

```

Python & Java : ['Arjun', 'Priya']
Python only : ['Karan']
All unique : {'Vikram', 'Arjun', 'Sneha', 'Divya', 'Karan', 'Neha', 'Priya', 'Rahul'}
Only one course: 3

```

```

In [2]: sentence = "the cat sat on the mat the cat sat"
        s=[]
        r=set()
        d={}
        s=sentence.split()
        l=len(max(s))
        for i in s:
            d.setdefault(i,s.count(i))
            if s.count(i)>1:
                r.add(i)
        print('Frequency',d)
        print('Words :',s)
        print('Most fequent :',max(s),l)
        print('Repeated words',r)

```

```

Frequency {'the': 3, 'cat': 2, 'sat': 2, 'on': 1, 'mat': 1}
Words : ['the', 'cat', 'sat', 'on', 'the', 'mat', 'the', 'cat', 'sat']
Most fequent : the 3
Repeated words {'sat', 'the', 'cat'}

```

In [ ]: